

GYM WORKOUT TYPE CLASSIFICATION (ML)

A supervised machine learning project predicting workout type from gym members' data

Agenda

01

Project Overview

Goal: predict workout type from member data

02

Dataset

10,000 members, 15 features

03

Pipeline

Cleaning, encoding, scaling, splitting

04

Models & Tuning

Logistic Regression, KNN, Decision Tree, GMM

05

Results & Conclusion

Best model, accuracy, next steps

Project Overview

This project builds a supervised classification model that predicts a gym member's workout type — Cardio, Strength, HIIT, or Yoga — using their demographic, biometric, and workout-session data. Several models are trained, compared, and the best one is tuned for optimal accuracy.

- Clean & prepare 10,000-row gym dataset
- Encode categorical features & scale numerics
- Train & compare 4 different models
- Tune hyperparameters for best accuracy

Project at a glance

10,000

gym member records

4

ML models trained & compared

68.4%

best model accuracy (tuned KNN)

The Dataset

gym_members_massive.csv · 10,000 rows × 15 columns

Demographics

Age, Gender

Body Metrics

Weight, Height, BMI, Fat %

Heart Rate

Max / Avg / Resting BPM

Workout Data

Duration, Frequency

Target Label

Workout_Type (4 classes)

Experience

Beginner to Advanced

ML Pipeline

1

Clean & Explore

Check nulls, types,
distributions, correlations

2

Encode

Label-encode Gender &
Workout_Type

3

Scale & Split

StandardScaler + 80/20
train-test split

4

Train Models

Logistic Regression, KNN,
Decision Tree, GMM

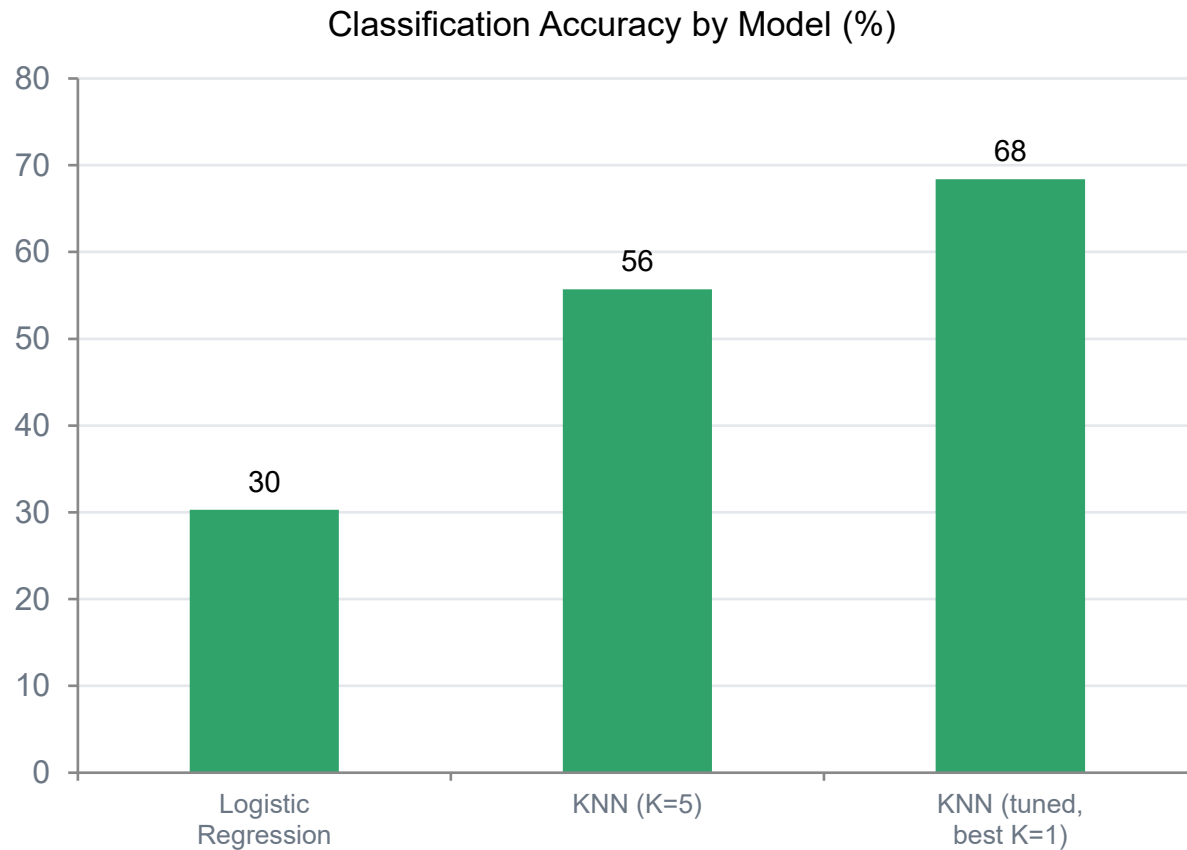
5

Tune & Evaluate

Best K search (1–20) +
accuracy/F1/confusion matrix

Tools: Python · Pandas · NumPy · Scikit-learn · Matplotlib · Seaborn · Jupyter Notebook

Models Trained & Compared



Logistic Regression

Baseline linear model — 30.3% accuracy

KNN (default K=5)

Distance-based model — 55.7% accuracy

KNN (tuned, K=1)

Best result after tuning — 68.4% accuracy

Decision Tree / GMM

Explored as regression & clustering alternatives

Conclusion & Next Steps

- The tuned K-Nearest Neighbors model reached 68.4% accuracy predicting workout type from biometric and session data — more than double the baseline logistic regression.
- This shows workout type carries a real, learnable signal in members' physiological and behavioral data, though room remains to improve further.

Next Steps

- Try ensemble models (Random Forest, XGBoost)
- Engineer new features (e.g. BPM ratios)
- Deploy the model behind a simple API

Thank you

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